Ground Then Grasp: Composing Part Grounding with a Language-Free Grasp Prior

Abstract

Language-driven grasp detection is a challenging task with applications in robotic manipulation and industrial automation. However, jointly localizing instruction-referred regions and predicting feasible grasps remains difficult. We propose Ground Then Grasp (G2G), a modular framework that decouples visual grounding from grasp prediction through spatial score fusion. A language-agnostic GR-ConvNet predicts grasp quality, orientation, and width maps, while CLIPSeg localizes the referred region. Both modules are trained independently on Grasp-Anything++. At inference, a weighted geometric mean fuses grounding and grasp quality scores, with reranking based on target-region coverage. We demonstrate that G2G produces an average successful detection on Grasp-Anything++ 37.7% (seen) and 35.8% (unseen). Code and models are available at and .

1. Introduction

Grasping is intuitive for humans but remains challenging for robots, where inappropriate contact points, such as a knife blade or a hot surface, may compromise safety and task execution. Language-driven grasp detection addresses this challenge by identifying grasp regions specified by natural-language instructions. However, jointly localizing the referred region and predicting feasible grasps remains difficult. We propose G2G, a modular framework that decouples visual grounding from grasp prediction.

2. Related Work

(i) Grasp Detection. GR-ConvNet [2] predicts dense grasp maps without language guidance. *(ii) Language-driven Grasping.* LGD [5] employs conditional diffusion, while subsequent methods explore mask-guided attention [4] and coarse-to-fine prediction [1]. *(iii) Visual Grounding.* CLIPSeg [3] provides text-conditioned segmentation. Unlike integrated grasping approaches.

Unlike previous works, our method independently trains grounding and grasp prediction modules and combines them through explicit spatial score fusion.

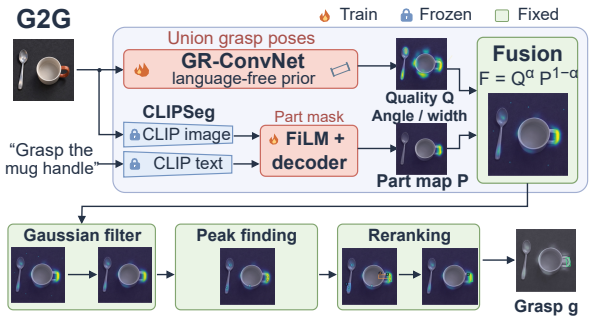


Figure 1. Overview of Ground Then Grasp, illustrating independent visual grounding and grasp prediction, score fusion, and coverage-based grasp reranking.

3. Method

Given an RGB image I and instruction t, G2G predicts a grasp rectangle $g = (x, y, w, h, \theta)$ in two stages: grounding-guided grasp scoring and grasp decoding and selection. We illustrate our pipeline in Fig. 1.

Grounding-Guided Grasp Scoring. *(i) Grasp poses prediction:* We adopt an RGB-based GR-ConvNet [2] to predict dense grasp quality q, orientation components (c, s), and normalized width u on a 224×224 grid. Its targets use the union of grasp poses from training-object annotations in each image (8,144 images and 9,862 objects); instruction is not used at this stage. For each annotated grasp, the central third of its rectangle is assigned a quality score of 1, while regions outside all annotated grasps are assigned 0. The four outputs are trained using summed mean Smooth L1 losses, with overlapping annotations resolved by retaining the last processed targets.

(ii) Language grounder: The grounder starts from pretrained CLIPSeg [3] (`clipseg-rd64-refined`). We freeze its CLIP encoders and fine-tune the decoder using binary cross-entropy with logits plus a soft Dice loss against the instruction's part mask. Grounding logits are bilinearly resized to the grasp grid.

The two modules are trained independently, without joint

loss, and combined only at inference.

Given the predictions from these two independently trained modules, we fuse grasp quality and part relevance into a unified spatial score map. Specifically, at each location p, let $Q(p) = \mathrm{clip}(q(p), 0, 1)$ denote grasp quality and $P(p) = \mathrm{sigmoid}(z(p)/\tau)$ denote grounding confidence, where $\tau > 0$ is a temperature parameter. We compute the fused score using a weighted geometric mean:

$$F(p) = Q(p)^{\alpha} P(p)^{1-\alpha}, \qquad 0 \le \alpha \le 1. \tag{1}$$

Here, α controls the relative contributions of grasp quality and grounding confidence.

Grasp Decoding and Selection. We apply Gaussian smoothing to all four maps (F, c, s, u). From the smoothed score $\bar{F}$, we extract up to 10 local maxima separated by at least 5 pixels. At each peak, $\theta = \frac{1}{2}\,\mathrm{atan2}(\bar{s}, \bar{c})$, $w = 112\,\mathrm{clip}(\bar{u}, 0, 1)$, and $h = w/2$. We rerank candidates by fused grasp quality and part coverage:

$$r_k = \log \max(\bar{F}(p_k), \epsilon) + \lambda \log \max(\bar{P}_k, \epsilon), \tag{2}$$

where $\bar{P}_k = \mathrm{mean}_{p \in \mathcal{R}_k} P(p)$, with $\mathcal{R}_k$ denoting pixels inside the grasp rectangle defined by center (x_k, y_k), orientation θ_k, in half-sized $(w_k/2, h_k/2)$ and $\epsilon = 10^{-6}$ avoids $\log 0$. The highest-ranked valid grasp is rescaled to the original image; if none exists, detection fails.

4. Experiments

Setting. We use dataset Grasp-Anything++ [5] with the defined base and new labels. We also follow the success rate metric, where a predicted grasp is correct if its IoU with an instruction-specific ground-truth grasp exceeds 25% and its angle offset is at most 30°. We train on 8,144 images (27,870 instructions) and test on 412 Seen images (1,376 instructions) and 193 Unseen images (601 instructions); 200 images (662 instructions) are held out for validation. We report seen (S), unseen (U), and harmonic mean H.

Baselines. We compare G2G with variants that share our prior and test manifests (Table 1); they differ only in how the instruction is used. *Prior only* ignores it. *Mask filter* thresholds the part mask (P) predicted by CLIPSeg (zero-shot or fine-tuned), sets the prior's grasp scores (Q) to zero outside it, and takes the best remaining grasp. *Mask peak* places the grasp at the most confident point of the predicted mask and reads the angle and width from the prior there. In contrast, G2G fuses the two scores softly, so a good grasp near an imperfect mask boundary is not discarded. *Oracle mask* replaces the predicted mask with the ground label for diagnostic. Due to limited time, we do not retrain other baselines but report two results published in [5].

Main Results. G2G improves on the prior by 4.0 (seen) and 3.5 (unseen) and on the fine-tuned mask filter by 1.2 and 2.3. However, grasp success with ground-truth masks reaches only 46.6/41.6%, suggesting that accurate localization alone does not guarantee successful grasp detection. G2G matches the reported GR-ConvNet + CLIP [5] on seen (37.6 vs. 37) but is far higher on unseen (35.8 vs. 18). We hypothesize that its jointly trained language branch is tied to training categories, whereas our language-free prior and pretrained CLIPSeg are not; the protocols differ and we did not test this. LGD + CLIP [5] remains the best-performing method (We report the results published by authors).

Variant	Seen	Unseen	H
Prior only	33.6	32.3	32.9
Mask filter (zero-shot)	32.0	33.3	32.7
Mask filter (fine-tuned)	36.4	33.4	34.9
Mask peak	18.4	13.8	15.8
G2G	**37.6**	**35.8**	**36.7**
Oracle mask	46.6	41.6	44.0
GR-ConvNet + CLIP [5]	37	18	24
LGD + CLIP [5]	48	42	45

Table 1. Top-1 grasp success (%), one seed.

We also measure pointing accuracy, defined as the percentage of grasp centers within the target mask. Fusion improves this from 31.2/40.3% to 61.0/66.4% (seen/unseen).

5. Conclusion

We presented G2G, a modular framework that combines independent visual grounding and grasp prediction through spatial score fusion. Experiments on Grasp-Anything++ demonstrate improvements over the shared grasp prior. Through this QACI project, I gained practical insights into robotic grasp detection. Future work will explore more robust grounding and flexible grasp representations to improve detection accuracy.

References

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